Week 2: Ordinary Differential Equations

CCS4004 - Optimization Techniques

Amgad A. Salama, Ph.D.

AAST - CCIT

Today's Journey

1. Why ODEs in Optimization?
2. First-Order ODEs
3. Systems of ODEs
4. Epidemic Modeling (SIR Model)
5. Applications: Electrical Circuits
6. Numerical Methods for ODEs
7. Optimal Control Problems
8. Let's Practice!
9. Summary & Applications

Quick Question

How do we model things that change over time?

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- Population growth? ✓
- Epidemic spread? ✓
- Chemical reactions? ✓
- Your bank account! ✓

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Answer: Differential Equations!

ODEs That Changed the World

1. **Lotka-Volterra (1920s)** - Predator-prey models
   - Enabled sustainable fishing quotas
   - Conservation biology applications
2. **SIR Model (1927)** - Epidemic theory
   - Guided COVID-19 response
   - Vaccination strategy optimization
3. **Rocket Equations (1960s)** - Space exploration
   - Moon landing trajectories
   - Modern SpaceX reusable rockets
4. **Circuit Theory (1900s)** - Electronics revolution
   - Every device you use!
   - Computer chip design

Did You Know?

Newton's Original Work

Isaac Newton invented calculus SPECIFICALLY to solve the ODE of planetary motion!

Weather Prediction

Weather forecasts solve systems of 100+ coupled ODEs in real-time!

Chaos Theory

The famous "butterfly effect" comes from studying sensitive dependence on initial conditions in ODEs.

COVID-19 Success

SIR models accurately predicted pandemic waves 2-3 weeks in advance, saving millions of lives.

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Why Study ODEs in Optimization?

The Connection

Many optimization problems involve dynamic systems that change over time.

Real-World Examples:

- **Optimal Control:** SpaceX rocket trajectory
- **Resource Management:** Sustainable fishing quotas
- **Drug Delivery:** Optimal medication dosing
- **Economics:** Optimal investment strategies over time

ODEs describe HOW systems evolve; Optimization finds the BEST evolution!

What is an Ordinary Differential Equation?

Definition

An **ODE** is an equation involving a function and its derivatives with respect to a single variable (usually time).

General Form:

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0$$

Simple Example:

$$\frac{dy}{dt} = ky$$

This describes exponential growth/decay!

- $k > 0$: Population growth
- $k < 0$: Radioactive decay

ODE vs Regular Equations

Algebraic Equation:

- Solve for a number
- Example: $2x + 3 = 7$
- Solution: $x = 2$

Differential Equation:

- Solve for a function
- Example: $\frac{dy}{dt} = 2y$
- Solution: $y(t) = Ce^{2t}$

Static Problem

Find a specific value

Dynamic Problem

Find how system evolves over time

ODEs tell us the STORY of how things change!

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First-Order ODEs

Definition

A first-order ODE involves only the first derivative:

$$\frac{dy}{dt} = f(t, y)$$

Common Types:

1. **Separable:** $\frac{dy}{dt} = g(t)h(y)$
2. **Linear:** $\frac{dy}{dt} + P(t)y = Q(t)$
3. **Exact:** $M(x, y) + N(x, y)\frac{dy}{dx} = 0$

Initial Condition

Need $y(t_0) = y_0$ to get unique solution!

Example: Population Growth

Problem: A bacteria population doubles every hour.

Model:

$$\frac{dP}{dt} = rP, \quad P(0) = P_0$$

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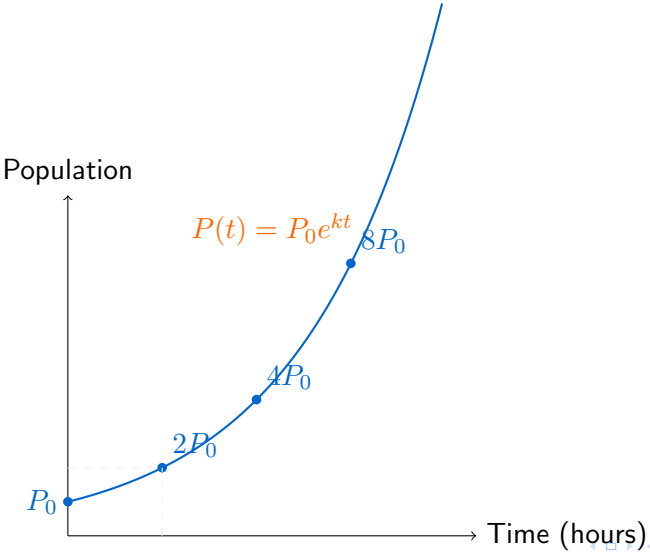
Solution Process:

$$\begin{aligned} \frac{dP}{P} &= r\,dt \\ \int \frac{dP}{P} &= \int r\,dt \\ \ln|P| &= rt + C \\ P(t) &= P_0 e^{rt} \end{aligned}$$

Doubling time: If $P(1) = 2P_0$, then $r = \ln(2) \approx 0.693$

Population Growth Visualization



Logistic Growth (With Constraints!)

Problem: Real populations have limited resources.

Model:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

where:

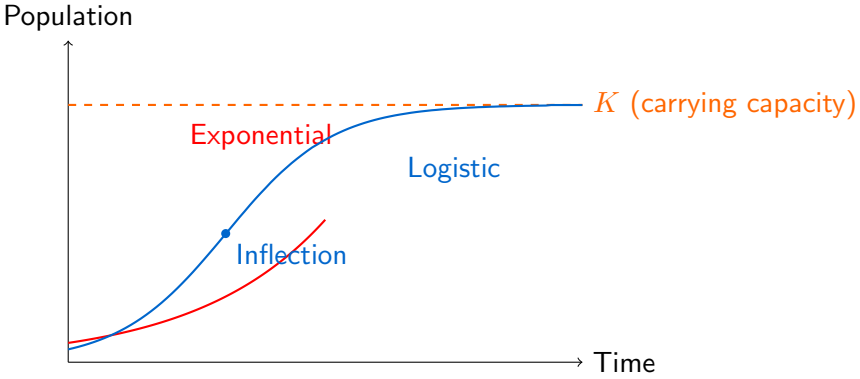
- r = growth rate
- K = carrying capacity (maximum sustainable population)

Behavior:

- When $P \ll K$: Growth is approximately exponential
- When $P \approx K$: Growth slows down
- When $P = K$: No growth (equilibrium)

This is a CONSTRAINED optimization problem in nature!

Logistic vs Exponential Growth



Real-World: Fish populations, bacterial cultures, product adoption

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Systems of ODEs

Definition

Multiple coupled ODEs that interact with each other:

$$\begin{cases} \frac{dx}{dt} = f(t, x, y) \\ \frac{dy}{dt} = g(t, x, y) \end{cases}$$

Why Important?

- Real systems have multiple interacting components
- Predator-prey dynamics (foxes and rabbits)
- Chemical reactions (multiple substances)
- Epidemics (susceptible, infected, recovered)

Vector Notation:

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y})$$

Example: Predator-Prey (Lotka-Volterra)

Scenario: Foxes eat rabbits in a forest.

Model:

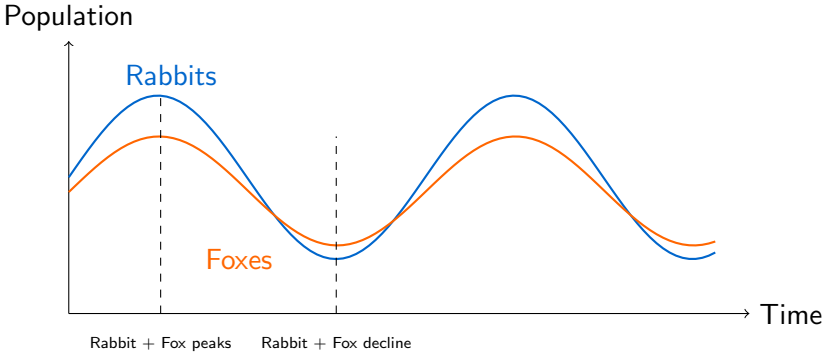
$$\begin{cases} \frac{dR}{dt} = aR - bRF & \text{(Rabbits)} \\ \frac{dF}{dt} = -cF + dRF & \text{(Foxes)} \end{cases}$$

Interpretation:

- aR: Rabbits reproduce naturally
- $-bRF$: Foxes eat rabbits (negative for rabbits)
- $-cF$: Foxes die without food
- $+dRF$: Foxes gain energy from eating (positive for foxes)

Result: Cyclical populations!

Predator-Prey Population Cycles



Pattern: Rabbit peak $\rightarrow$ Fox peak follows $\rightarrow$ Rabbit decline $\rightarrow$ Fox decline $\rightarrow$ Repeat

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SIR Epidemic Model

Model COVID-19, flu, measles, etc.

Three Groups:

- **S**usceptible: Can catch disease
- **I**nfected: Currently sick, can spread disease
- **R**ecovered: Immune, can't spread or catch again

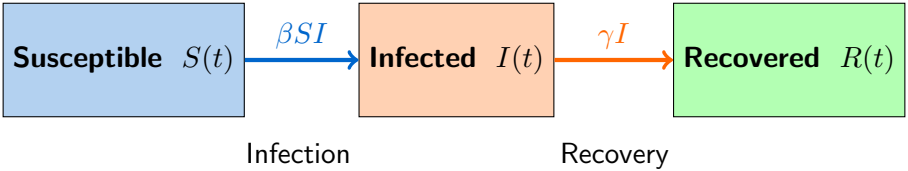
SIR Equations:

$$\begin{cases} \frac{dS}{dt} = -\beta SI \\ \frac{dI}{dt} = \beta SI - \gamma I \\ \frac{dR}{dt} = \gamma I \end{cases}$$

where:

- β = transmission rate (contact $\times$ infection probability)
- γ = recovery rate (inverse of infectious period)

SIR Flow Diagram

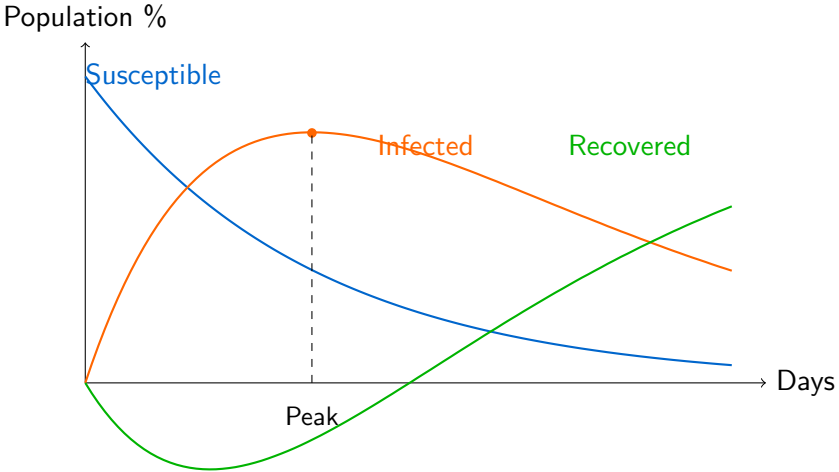


Key Quantity: Basic Reproduction Number

$$R_0 = \frac{\beta}{\gamma}$$

- $R_0 > 1$: Epidemic spreads
- $R_0 < 1$: Epidemic dies out
- COVID-19: $R_0 \approx 2 - 3$

SIR Model Simulation



Optimization Problem: How to minimize peak infections? (Flatten the curve!)

Real World: COVID-19 Modeling

How SIR models guided pandemic response:

Questions Answered:

- When will cases peak?
- How many hospital beds needed?
- Effect of lockdowns?
- Vaccination strategy?

Optimization Goals:

- Minimize deaths
- Avoid hospital overload
- Minimize economic impact

Control Variables:

- Social distancing ($\downarrow \beta$)
- Testing and isolation
- Vaccination rate
- Mask mandates

Result

SIR models helped save millions of lives by informing policy decisions!

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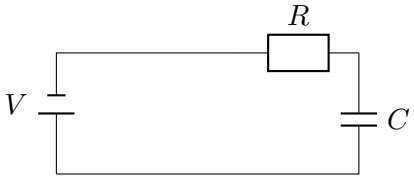
Application: RC Circuits

Problem: Charging a capacitor through a resistor.

Circuit Diagram:



ODE:

$$RC\frac{dV_C}{dt} + V_C = V$$

Solution:

$$V_C(t) = V(1 - e^{-t/RC})$$

Time Constant: $\tau = RC$

- At $t = \tau$: 63% charged
- At $t = 5\tau$: 99% charged

Optimizing Circuit Design

Design Challenge: Choose R and C for optimal performance.

Competing Objectives:

- **Fast charging:** Small RC (small time constant)
- **Energy efficiency:** Minimize power dissipation I^2R
- **Cost:** Cheaper components have limited values
- **Space:** Physical size constraints

Optimization Problem:

$$\begin{aligned} \min \quad & \alpha \cdot RC + \beta \cdot \text{Cost}(R, C) \\ \text{s.t.} \quad & V_C(t_{\text{target}}) \geq 0.95V \\ & R_{\min} \leq R \leq R_{\max} \\ & C_{\min} \leq C \leq C_{\max} \end{aligned}$$

ODEs provide the constraints for optimization!

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Why Numerical Methods?

The Problem

Most ODEs cannot be solved analytically!

Examples with no closed-form solution:

- $\frac{dy}{dt} = y^2 + t^2$
- Nonlinear systems (SIR, predator-prey)
- Most real-world problems!

Solution: Use numerical approximation!

Key Numerical Methods:

1. Euler's Method (simplest)
2. Runge-Kutta Methods (most popular)
3. Adams Methods (multistep)

Euler's Method

Idea: Approximate curve with small straight-line segments.

Given: $\frac{dy}{dt} = f(t, y)$, $y(t_0) = y_0$

Algorithm:

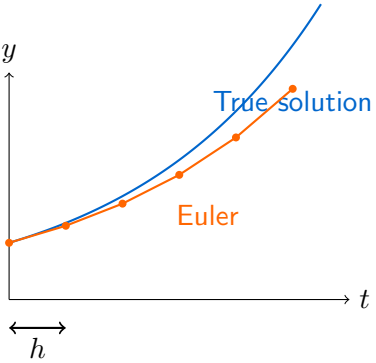
1. Choose step size h
2. For each step: $y_{n+1} = y_n + h \cdot f(t_n, y_n)$
3. Update: $t_{n+1} = t_n + h$

Example: $\frac{dy}{dt} = y$, $y(0) = 1$, $h = 0.1$

$$
\begin{aligned}
y_1 &= y_0 + h \cdot y_0 = 1 + 0.1(1) = 1.1 \\
y_2 &= y_1 + h \cdot y_1 = 1.1 + 0.1(1.1) = 1.21 \\
&\vdots
\end{aligned}
$$

Exact: $y(t) = e^t$, so $y(0.1) = e^{0.1} \approx 1.105$

Euler's Method Visualization



Key Points:

- Smaller h = better accuracy
- Accumulates error over many steps
- Simple but not very accurate

Runge-Kutta Method (RK4)

The industry standard for ODE solving!

Idea: Use multiple slope evaluations for better accuracy.

RK4 Algorithm:

$$
\begin{aligned}
k_1 &= f(t_n, y_n) \\
k_2 &= f(t_n + h/2, y_n + hk_1/2) \\
k_3 &= f(t_n + h/2, y_n + hk_2/2) \\
k_4 &= f(t_n + h, y_n + hk_3) \\
y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}
$$

Advantages:

- 4th-order accuracy (error $\sim h^4$)
- Much better than Euler (error $\sim h$)
- Used in SciPy, MATLAB, industry software

Python Implementation

Solving ODEs with SciPy:

```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

# Define the ODE: dy/dt = -k*y (exponential decay)
def model(y, t, k):
    dydt = -k * y
    return dydt

# Parameters
y0 = 100          # Initial population
k = 0.5           # Decay rate
t = np.linspace(0, 10, 100)  # Time points

# Solve ODE
y = odeint(model, y0, t, args=(k,))

# Plot
plt.plot(t, y, 'b-', label='Numerical␣Solution')
plt.plot(t, y0*np.exp(-k*t), 'r--', label='Analytical')
plt.xlabel('Time')
plt.ylabel('Population')
plt.legend()
plt.show()
```

Run this code yourself!

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Optimal Control: The Connection to Optimization

Optimal Control Problem

Find the control function $u(t)$ that minimizes/maximizes an objective while satisfying ODE constraints.

General Form:

$$\begin{aligned} \min_{u(t)} \quad & J = \int_{t_0}^{t_f} L(x(t), u(t), t)\, dt \\ \text{s.t.} \quad & \frac{dx}{dt} = f(x(t), u(t), t) \\ & x(t_0) = x_0, \quad x(t_f) = x_f \end{aligned}$$

Components:

- $x(t)$ = state variables (governed by ODEs)
- $u(t)$ = control variables (what we choose)
- J = objective functional (what we optimize)

Example: Minimum Time Problem

Scenario: Move a robot from point A to B in minimum time.

State Equations (ODEs):

$$\begin{cases} \frac{dx}{dt} = v \\ \frac{dv}{dt} = u \end{cases}$$

where:

- $x =$ position
- $v =$ velocity
- $u =$ acceleration (control)

Objective:

$$\min_{u(t)} \quad t_f \quad \text{(minimize final time)}$$

Constraints:

- $x(0) = 0$, $x(t_f) = L$ (start and end positions)
- $|u| \leq u_{\max}$ (acceleration limits)

Real World: SpaceX Falcon 9 Landing

Real World: SpaceX Falcon 9 Landing

One of the most complex optimal control problems solved!

State Variables (ODEs):

- Position: $\mathbf{r}(t)$
- Velocity: $\mathbf{v}(t)$
- Mass: $m(t)$ (fuel burning)
- Attitude: $\theta(t)$ (orientation)

Control Variables:

- Thrust magnitude
- Thrust direction
- Grid fin angles

Objective:

$$\min \int_0^{t_f} |\mathbf{F}(t)|\, dt$$

(minimize fuel consumption)

Constraints:

- Physics (gravity, drag)
- $\mathbf{r}(t_f) = \mathbf{r}_{\text{target}}$
- $\mathbf{v}(t_f) \approx 0$
- Maximum thrust limits
- Real-time computation

Application: Optimal Drug Dosing

Problem: Deliver medication to maintain therapeutic levels.

Pharmacokinetic Model (ODE):

$$\frac{dC}{dt} = u(t) - kC$$

where:

- $C(t)$ = drug concentration in blood
- $u(t)$ = dosing rate (control)
- k = elimination rate

Optimization Goal:

$$\begin{aligned} \min \quad & \int_0^T (C(t) - C_{\text{target}})^2 \, dt + \lambda \int_0^T u(t)^2 \, dt \\ \text{s.t.} \quad & C_{\min} \leq C(t) \leq C_{\max} \\ & u(t) \geq 0 \end{aligned}$$

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Cooling Coffee Problem

Scenario: Your coffee starts at 90C in a 20C room.

Newton's Law of Cooling:

$$\frac{dT}{dt} = -k(T - T_{\text{room}})$$

Your Tasks (3 minutes):

1. What type of ODE is this? (Order? Linear?)
2. If coffee cools to 70C in 5 minutes, what is k?
3. When will coffee reach 40C (drinkable)?

Work with your neighbor!

Cooling Coffee - Solution

1. **Type:** First-order linear ODE
2. **Finding k:**

$$T(t) = T_{\text{room}} + (T_0 - T_{\text{room}})e^{-kt}$$

Given: $T_0 = 90$, $T(5) = 70$, $T_{\text{room}} = 20$

$$70 = 20 + (90 - 20)e^{-5k} \implies e^{-5k} = \frac{50}{70} \implies k \approx 0.0673$$

3. **When $T = 40$?**

$$40 = 20 + 70e^{-0.0673t} \implies t \approx 17.3 \text{ minutes}$$

Your coffee is drinkable in about 17 minutes!

Quick Exercise: Bacteria Growth

A bacteria population doubles every 30 minutes.

Starting with 1000 bacteria, answer:

1. Write the ODE

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   $P(0.5) = 2P(0) \implies 2000 = 1000e^{0.5k} \implies k = 2\ln(2) \approx 1.386$

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3. How many bacteria after 2 hours?
   $P(2) = 1000e^{2 \cdot 1.386} = 1000e^{2.772} \approx 16,000$ bacteria

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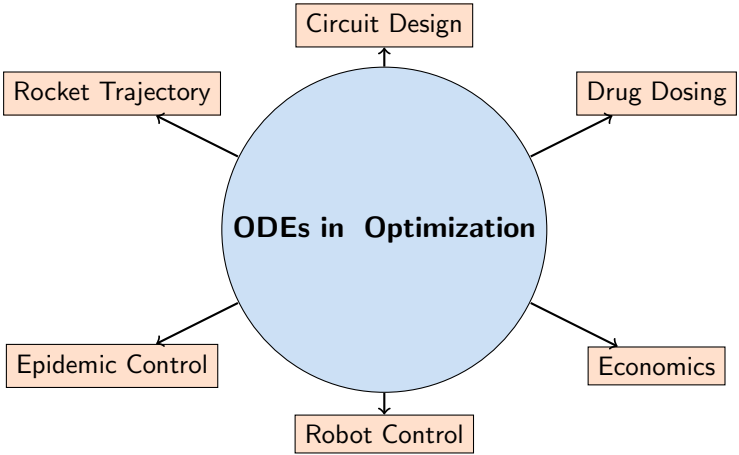
ODEs in Optimization: The Big Picture



Today's Key Takeaways

1. **ODEs model dynamic systems** - Things that change over time
2. **Everywhere in real world** - Populations, epidemics, circuits, rockets
3. **Types of ODEs** - First-order, systems, linear/nonlinear
4. **Numerical methods** - Euler, Runge-Kutta for unsolvable ODEs
5. **Optimal control** - ODEs as constraints in optimization
6. **Python tools** - SciPy makes it easy!

ODEs + Optimization = Powerful combination!

This Week's Tasks

Assignment 2 (Due next week):

1. Solve 3 ODEs analytically (provided in assignment sheet)
2. Implement Euler's method in Python for one problem
3. Simulate SIR model with different β values
4. Write short report: How does R_0 affect epidemic?

Practice:

- Install SciPy if you haven't already
- Try modifying the Python code from today's lecture

Thank You!

See you next week!